Point Estimation:

Recall:

+ Context: $X_1, \ldots X_n$ are iid random variables, with θ being some unknown parameter of interest. (E.g. $X_1, \ldots X_n \overset{iid}{\sim} Exp(\theta)$, or $X_1, \ldots X_n$ iid w/ $\theta = E[X_1]$)
+ Goal: Find a "good" way to estimate the unknown parameter θ.

Def. (Point Estimator): Given a random sample $X_1, \ldots X_n$, a point estimator for θ, denoted $\hat{\theta}(X_1, \ldots X_n)$, is any real-valued function of $X_1, \ldots X_n$.

Examples:

+ $\hat{\theta}(X_1, \ldots X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i$ ← The (random) mean of a sample
+ $\hat{\theta}(X_1, \ldots X_n) = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \frac{1}{n}\sum_{j=1}^{n} X_j\right)^2$ ← The (random) variance of a sample

Note: Since $X_1, \ldots X_n$ are random, then the point estimator $\hat{\theta}(X_1, \ldots X_n)$ is also random.

Note: A statistic and a point estimator differ only in that a point estimator gives us a "guess" for an underlying parameter θ. A statistic need not do so.

Once we actually perform sampling, we will obtain realizations $x_1, \ldots x_n$, and hence, a realization of the point estimator, given by $\hat{\theta}(x_1, \ldots x_n)$.

A realization of a point estimat<u>or</u> is called a point estimat<u>e</u>.

Pre-Sampling	Post-Sampling
R.V.'s $X_1, \ldots X_n$	Realizations $x_1, \ldots x_n$
Estimat<u>or</u> $\hat{\theta}(X_1, \ldots X_n)$	Estimat<u>e</u> $\hat{\theta}(x_1, \ldots x_n)$

We will analyze properties of different point estimators, whereas a point estimate can be regarded as a single "sensible" value of the unknown parameter θ.

Since point estimators are random, we need to study their distribution and think about how to assess their quality.

$\rightsquigarrow$ We want to look into the sampling distribution of different point estimators

Motivating Question:

What makes a point estimator "good" or "bad"?

Ex: Estimating the mean of a Normally distributed population.

Let $X_1, X_2 \overset{iid}{\sim} N(\theta, 1)$, where $\theta \in \mathbb{R}$ is unknown.

Consider three point estimators for θ:

+ $\hat{\theta}_1(X_1, X_2) = 0$
+ $\hat{\theta}_2(X_1, X_2) = X_1 - X_2$.
+ $\hat{\theta}_3(X_1, X_2) = \frac{1}{2}(X_1 + X_2)$

Ideally, we want the distribution of a point estimator to depend on the true value of the parameter θ.

Otherwise, the distribution of $\hat{\theta}(X_1, X_2)$ doesn't really let us differentiate between possible values of θ.

Looking at our three point estimators:

+ $\hat{\theta}_1$ is always constant and equal to zero – regardless of the value of θ.

$\Rightarrow$ $\hat{\theta}_1$ is not a good point estimator – its sampling distribution contains no information about the actual value of θ

+ $\hat{\theta}_2(X_1, X_2) = X_1 - X_2$. Since $X_1, X_2 \overset{iid}{\sim} N(\theta, 1)$, then:

$\hat{\theta}_2(X_1, X_2) \sim N(\theta - \theta, 1+1) = N(0, 2)$. Still, this doesn't depend on θ in any way.

$\Rightarrow$ $\hat{\theta}_2$ is not a good point estimator – similar issue to $\hat{\theta}_1$

+ $\hat{\theta}_3(X_1, X_2) = \frac{1}{2}(X_1 + X_2)$. Same as above, we see that:

$\hat{\theta}_3(X_1, X_2) \sim N\left(\frac{1}{2}(\theta+\theta), \frac{1}{4}(1+1)\right) = N\left(\boxed{\theta}, \frac{1}{2}\right)$. This does depend on θ!

Varying θ $\Rightarrow$ Different Dist'n of $\hat{\theta}_3(\dots)$!

$\Rightarrow$ $\hat{\theta}_3$ seems like a good point estimator, since its distribution is affected by θ.

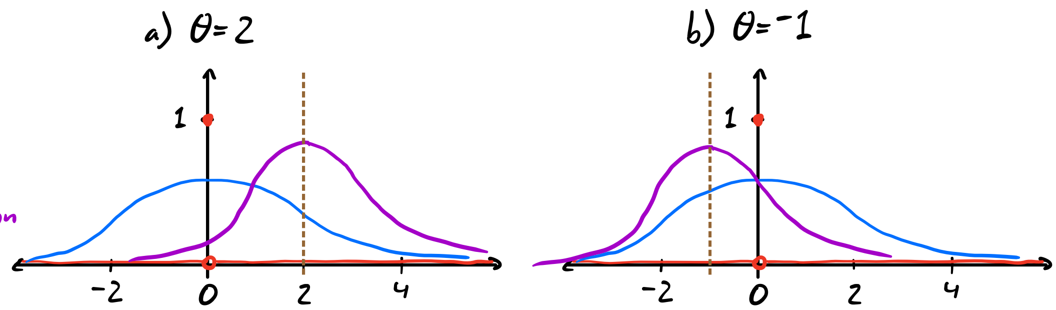
Illustration: Sampling Distributions for two possible values of θ

: pmf of $\hat{\theta}_1$

: pdf of $\hat{\theta}_2$

: pdf of $\hat{\theta}_3$

$\Rightarrow$ Only the distribution of $\hat{\theta}_3$ varies w/ θ



Summary:

Being given the distribution of either $\hat{\theta}_1(\dots)$ or $\hat{\theta}_2(\dots)$ yields no information about the unknown parameter θ.

But, if we are given the distribution of $\hat{\theta}_3(\dots)$, then we can find θ! It is the mean of $\hat{\theta}_3(\dots)$!

1st Principle of Point Estimation: Informativeness

We should always choose a point estimator (for θ) whose sampling distribution is affected by the true value of θ.

Next: What other properties might we want the sampling distribution of a point estimator to have?

(Un-)Biased Estimators:

We may want the distribution of $\hat{\theta}(X_1 \ldots X_n)$ to be "centered at" the value of the unknown θ.

Def. (Unbiased, Bias): A point estimator $\hat{\theta}(X_1, \ldots X_n)$ is an *unbiased estimator for θ* if $\mathbb{E}[\hat{\theta}(X_1 \ldots X_n)] = \theta$.

If not, we define the *bias of $\hat{\theta}(X_1 \ldots X_n)$ with respect to θ* as: $\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta}(X_1 \ldots X_n)] - \theta$.

A Very Important Note: The "true" value of the unknown parameter θ is not random! Even though it is unknown, it is not uncertain.

Example: Let $X_1 \ldots X_n \overset{iid}{\sim} N(\underbrace{\mu}_{``=\theta_1"}, \underbrace{\sigma^2}_{``=\theta_2"})$, where both $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ are unknown.

Claim 1: The sample mean $\hat{\mu}(X_1 \ldots X_n) = \frac{1}{n}\sum_{i=1}^n X_i$ is an unbiased estimator for μ.

Proof:
$$\mathbb{E}[\hat{\mu}(X_1 \ldots X_n)] = \mathbb{E}[\tfrac{1}{n}\textstyle\sum_{i=1}^n X_i]$$
Linearity of $\mathbb{E}[\cdot]$
$$= \tfrac{1}{n}\textstyle\sum_{i=1}^n \mathbb{E}[X_i]$$
$X_1 \ldots X_n$ are iid, $\mathbb{E}[X_i] = \mu$ for all i
$$\mathbb{E}[\hat{\mu}(X_1 \ldots X_n)] = \tfrac{1}{n}\cdot\textstyle\sum_{i=1}^n \mu = \mu \quad \blacksquare$$

Claim 2: The sample variance $\hat{\sigma}^2(X_1 \ldots X_n) = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2$, where $\bar{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$, is a biased estimator for σ^2.

Proof:
$$\mathbb{E}[\hat{\sigma}^2(X_1 \ldots X_n)] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n (X_i - \bar{X}_n)^2\right]$$
Expand the square and distribute the summation
$$= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i^2 - 2\cdot\underbrace{\frac{1}{n}\sum_{i=1}^n X_i}_{=\bar{X}_n}\cdot\bar{X}_n + \underbrace{\frac{1}{n}\sum_{i=1}^n \bar{X}_n^2}_{=\frac{1}{n}\cdot n\cdot\bar{X}_n^2}\right]$$

$$= \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i^2\right] - \cancel{2}\,\mathbb{E}\left[\bar{X}_n^2\right] + \cancel{\mathbb{E}\left[\bar{X}_n^2\right]}$$ } Linearity of $\mathbb{E}[\cdot]$

} Linearity of $\mathbb{E}[\cdot]$

$$= \cancel{\tfrac{1}{n}}\cancel{\textstyle\sum_{i=1}^{n}}\,\mathbb{E}[X_i^2] - \mathbb{E}[\bar{X}_n^2]$$

Since $X_1 \dots X_n$ are iid } $\mathbb{E}[X_1^2] = \mathbb{E}[X_2^2] = \dots$

$$= \underbrace{\mathbb{E}[X_1^2]}_{= Var(X_1) - \mathbb{E}[X_1]^2,\ \text{by def'n of Variance}} - \underbrace{\mathbb{E}[\bar{X}_n^2]}_{= Var(\bar{X}_n) - \mathbb{E}[\bar{X}_n]^2,\ \text{by def'n of Variance}}$$

$$= \underbrace{Var(X_1)}_{=\sigma^2} - \underbrace{\mathbb{E}[X_1]}_{=\mu}^2 - \underbrace{Var(\bar{X}_n)}_{=\sigma^2/n} + \underbrace{\mathbb{E}[\bar{X}_n]}_{=\mu}^2$$

$$= \sigma^2 - \cancel{\mu^2} - \sigma^2/n + \cancel{\mu^2}$$

$$\underline{\mathbb{E}\left[\hat{\sigma}^2(X_1 \dots X_n)\right] = \frac{n-1}{n}\cdot\sigma^2} \neq \sigma^2$$

Hence, the bias of $\hat{\sigma}^2(X_1 \dots X_n)$ is:

$$\underline{Bias}(\hat{\sigma}^2, \sigma^2) = \frac{n-1}{n}\cdot\sigma^2 - \sigma^2 = \frac{-1}{n}\sigma^2$$

$\rightsquigarrow$ By using $\hat{\sigma}^2(X_1 \dots X_n)$ to estimate σ^2, we will tend to underestimate the true value of σ^2. $\boxtimes$

Note: In this example, the point estimator $\hat{s}^2(X_1 \dots X_n) = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ is an unbiased estimator of σ^2. [Try to show this as an exercise!]

This is why sometimes people will refer to $\hat{s}^2(X_1 \dots X_n)$ as the "sample variance", and to $\hat{\sigma}^2(X_1 \dots X_n)$ as the "population variance".

We will not use their terminology, but you certainly should be familiar with it.

Note: We showed that $\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2$ was a biased estimator for σ^2.

If we knew the true mean μ, we could compute $\frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$.

Would this second estimator for σ^2 also be biased? [An exercise!]

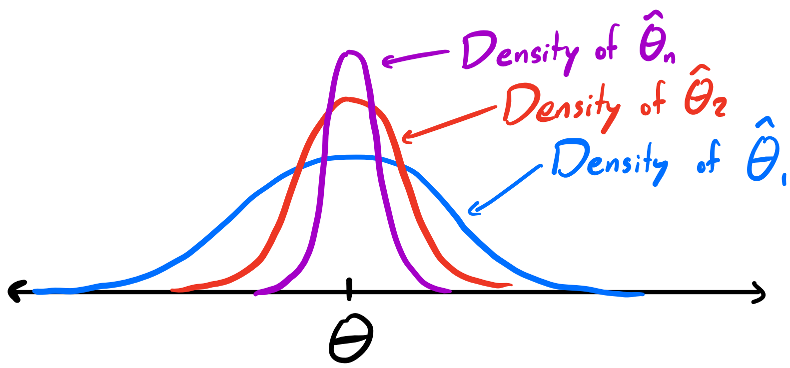
Q: Are unbiased estimators unique?

A: No! Let $X_1, \dots X_n \overset{iid}{\sim} N(\theta, 1)$, with unknown mean θ. Consider the following point estimators:

$$\left.\begin{array}{l} \bullet\ \hat{\theta}_1(X_1 \dots X_n) = X_1 \\ \bullet\ \hat{\theta}_2(X_1 \dots X_n) = \frac{1}{2}(X_1 + X_2) \\ \qquad \vdots \\ \bullet\ \hat{\theta}_n(X_1 \dots X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i \end{array}\right\} \text{All of these are unbiased estimators of } \theta!$$

If we have multiple unbiased estimators, how do we choose between them?

$\rightsquigarrow$ Look at their distributions!



$\hat{\theta}_n \sim N(\theta, \frac{1}{n})$

$\hat{\theta}_2 \sim N(\theta, \frac{1}{2})$

$\hat{\theta}_1 \sim N(\theta, 1)$

Notice: Since the variance of $\hat{\theta}_n$ is less than the variance of $\hat{\theta}_1$ and $\hat{\theta}_2$, then $\hat{\theta}_n$ is, on average, closer to its mean (θ) than of $\hat{\theta}_1$ and $\hat{\theta}_2$.

If the estimator $\hat{\theta}(X_1 \dots X_n)$ is unbiased and has low variance, then an estimate $\hat{\theta}(x_1 \dots x_n)$ is "likely" closer to θ.

$\Rightarrow$ Lower variance estimators are better than higher variance estimators!

As a convention, we refer to the standard deviation of a point estimator by a special name.

Def. (Standard Error): For a point estimator $\hat{\theta}(X_1 \dots X_n)$, its *standard error* is:

$$SE(\hat{\theta}) := \sqrt{Var(\hat{\theta}(X_1 \dots X_n))}$$

I.e., $SE(\hat{\theta})$ is the standard deviation of $\hat{\theta}(X_1 \dots X_n)$.

Note: Low $SE(\hat{\theta})$ means $\hat{\theta}(X_1 \dots X_n)$ is usually closer to $E[\hat{\theta}(X_1 \dots X_n)]$, but if $\hat{\theta}(\dots)$ is biased, then $E[\hat{\theta}(X_1 \dots X_n)] \neq \theta$.

$\rightsquigarrow$ So we want an estimator with both: i) low (or zero) bias, and ii) low Std. Error / Variance

Def. (Min. Variance Unbiased Estimator): We say that $\hat{\theta}(X_1 \dots X_n)$ is the *minimum variance unbiased estimator (MVUE)* for θ if both:

- $E[\hat{\theta}(X_1 \dots X_n)] = \theta$
- $Var(\hat{\theta}(X_1 \dots X_n)) \leq Var(\tilde{\theta}(X_1 \dots X_n))$ for all unbiased estimators $\tilde{\theta}(\dots)$

Note: The MVUE has the tightest possible distribution centered around the true value of θ. Hence, it is "best" in some sense.

Q: How do we know or prove that some estimator is the MVUE?

$\rightsquigarrow$ Take more statistics! (Beyond the scope of this class)

$$\boxed{\begin{array}{l}\text{2nd Principle of Point Estimation: Unbiasedness} \\ \quad \text{Choose unbiased estimators with small variance}\end{array}}$$

Quality of Estimators:

A common way to measure the quality of a point estimator is the following:

Def. (Mean Squared Error, MSE): Let $X_1, \dots X_n$ be iid from some population with unknown parameter θ. If $\hat{\theta}(x_1, \dots x_n)$ is a point estimator of θ, then the *mean squared error* of $\hat{\theta}(\dots)$ with respect to θ is:

$$MSE(\hat{\theta}, \theta) = \mathbb{E}\left[\left(\hat{\theta}(x_1, \dots x_n) - \theta\right)^2\right]$$

over the randomness of $X_1 \dots X_n$, i.e. the sampling distribution of $\hat{\theta}(X_1 \dots X_n)$

Intuition: MSE captures how far $\hat{\theta}(\dots)$ is from the true value of θ, on average.

Lower MSE $\rightsquigarrow$ Better Quality Point Estimator

Claim: $MSE(\hat{\theta}, \theta) = \underbrace{Bias(\hat{\theta}, \theta)^2}_{\text{"accuracy"}} + \underbrace{Var(\hat{\theta})}_{\text{"precision"}}$ // Captures both principles of point estimation!

Proof: Exercise.

Ex.: $X_1 \dots X_n \overset{iid}{\sim} Unif[0, 2\cdot\theta]$ for unknown θ

2 Candidate Point Estimators:

- $\hat{\theta}_1(x_1 \dots x_n) = 2 \cdot \frac{1}{n} \sum_{i=1}^{n} X_i$ (2× the mean)
- $\hat{\theta}_2(X_1 \dots X_n) = \max_i X_i$ (largest value of the $X_1 \dots X_n$)

Bias? Variance? MSE?

=> Which might we prefer and why? (TBD in Precepts)